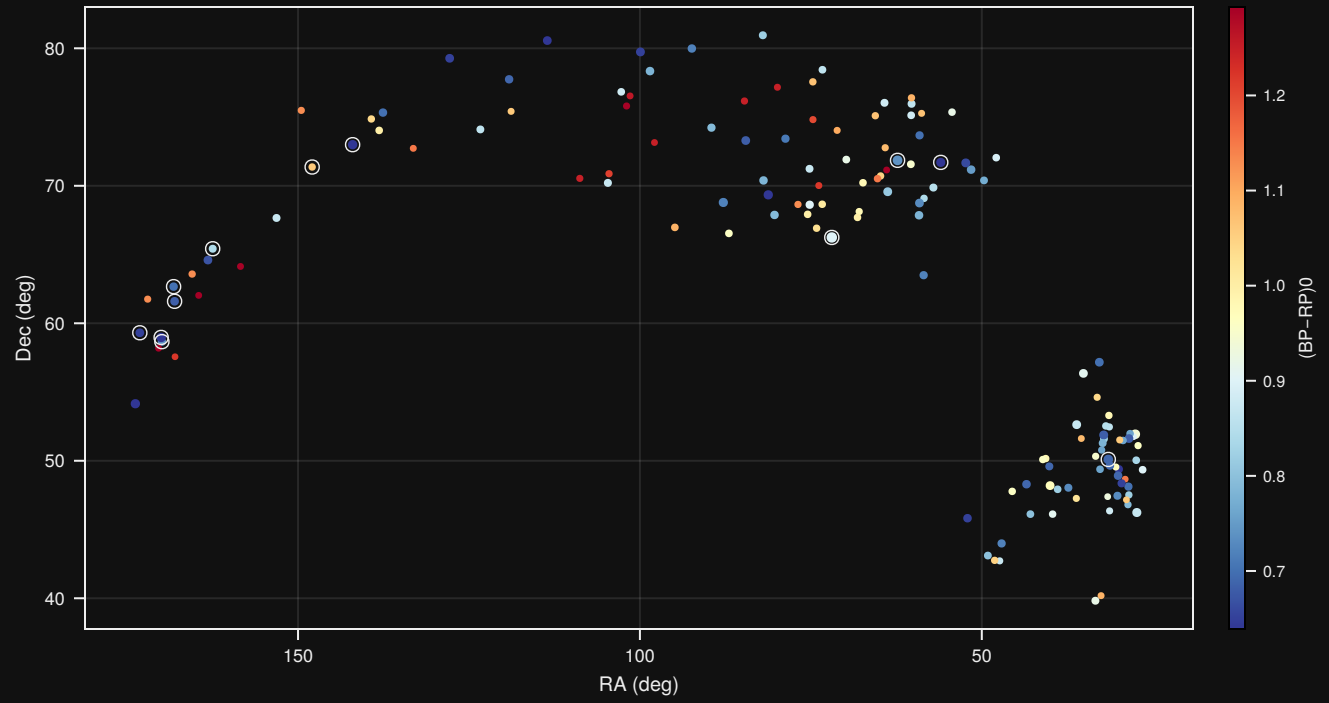
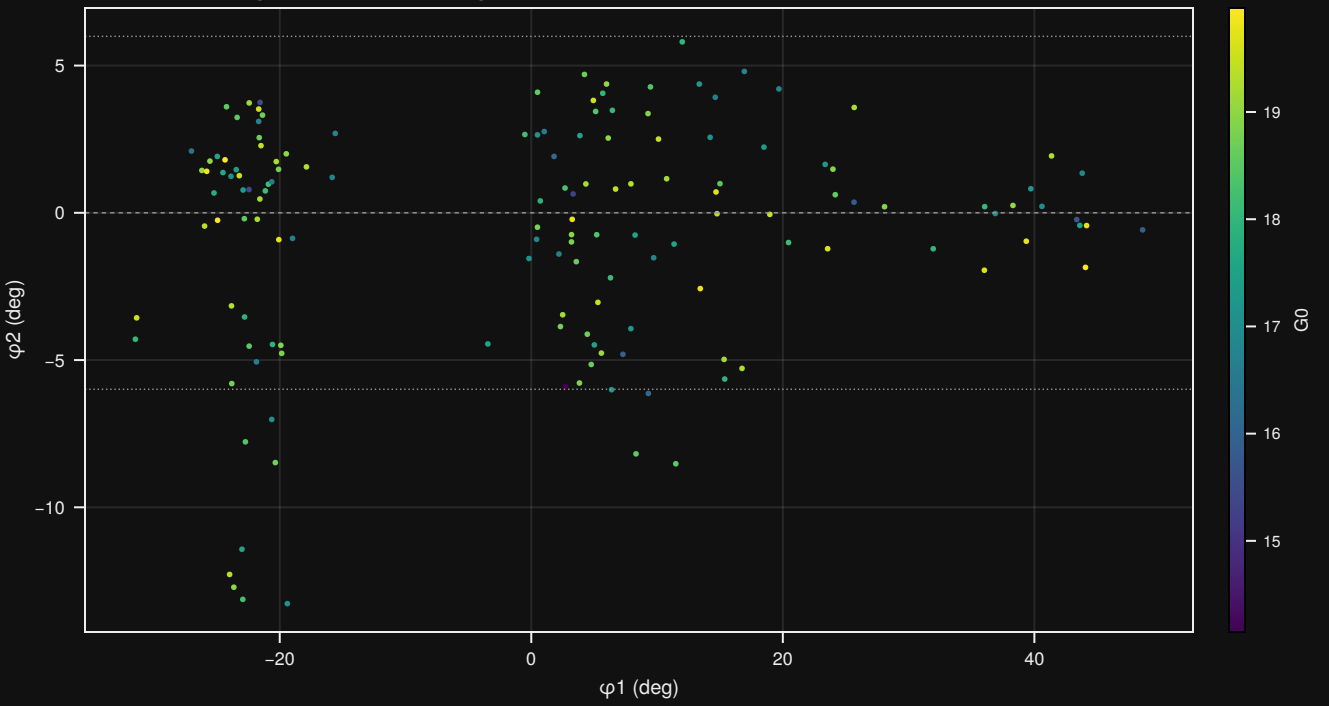
New-6

MEMBERSHIP	
Stars in catalog	146
with Gaia GSP-Phot	77 (53%)
with radial velocity	12
Stream id (sID)	22
GEOMETRY · great-circle fit	
Centroid RA, Dec	68.4, 66.7 deg
Centroid l, b	138.4, 14.5 deg
Pole RA, Dec	285.3, 14.1 deg
Length (φ1, 2–98%)	70.1 deg
Width σ(φ2), robust	3.00 deg
KINEMATICS	
pmRA* med ± σ	-4.58 ± 11.35 mas/yr
pmDE med ± σ	-15.51 ± 5.41 mas/yr
μφ1* med ± σ	-16.27 ± 3.36 mas/yr
μφ2 med ± σ	-12.41 ± 3.81 mas/yr
Parallax (cat.) med	0.313 ± 0.185 mas
Parallax (wtd, ZP-corr.)	0.353 ± 0.013 mas
Distance	d = 2.8 ± 0.1 kpc
Physical length	3.46 kpc
RV median ± σ(MAD)	-119.2 ± 43.9 km/s
RV gradient dV/dφ1	-2.84 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	14.2–20.0 (18.5)
(BP−RP)0 median	0.87
GSP-Phot [M/H] (n=77)	-2.26 ± 0.43
GSP-Phot Teff median	5258 K
RUWE > 1.4	1.3 %
PM clusters (largest)	8 (50 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	146 | 146
n_RV catalog | paper	12 | 12
<[Fe/H]> paper	-2.18
Stellar mass (paper)	340 Msun
Pole offset vs paper	1.6 deg

Sky map · catalog members (n = 146) · size ∝ brightness · ring = has RV

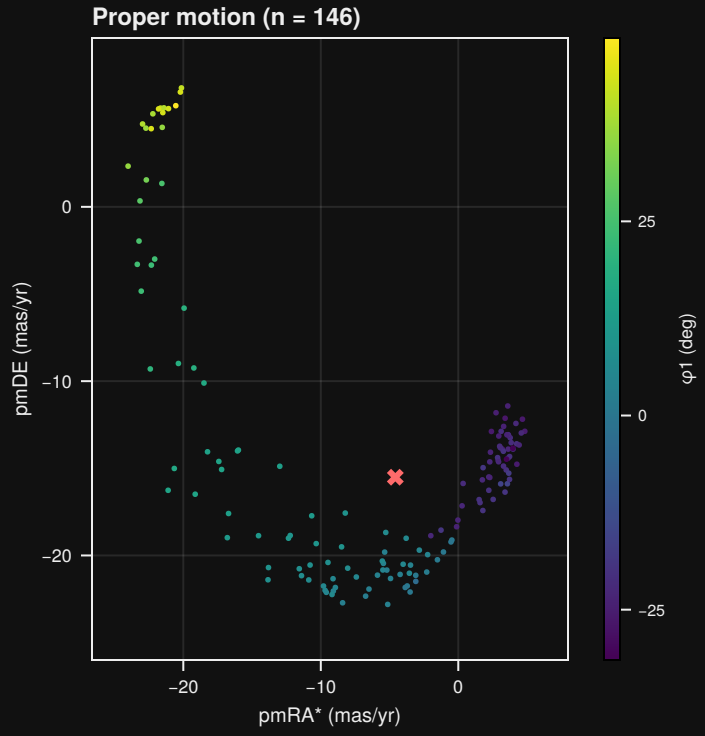


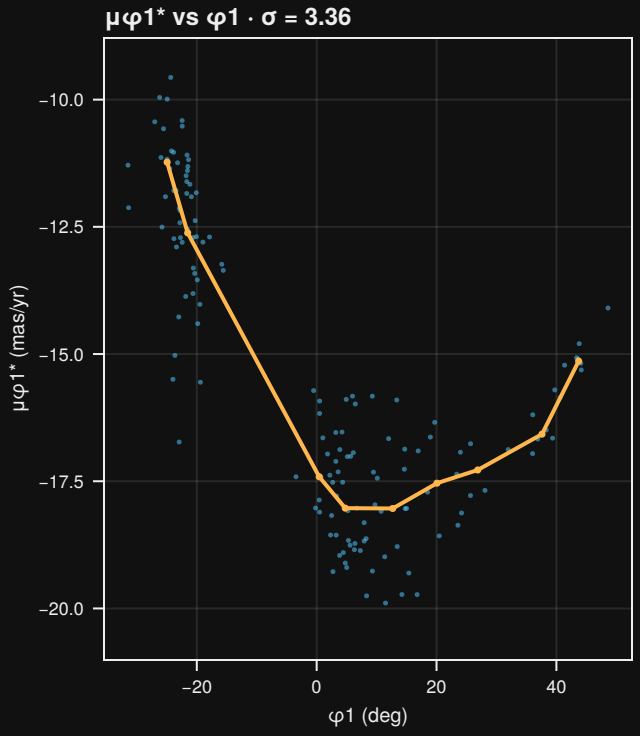
Stream frame (great-circle fit) · length 70.1° · width σ 3.00°

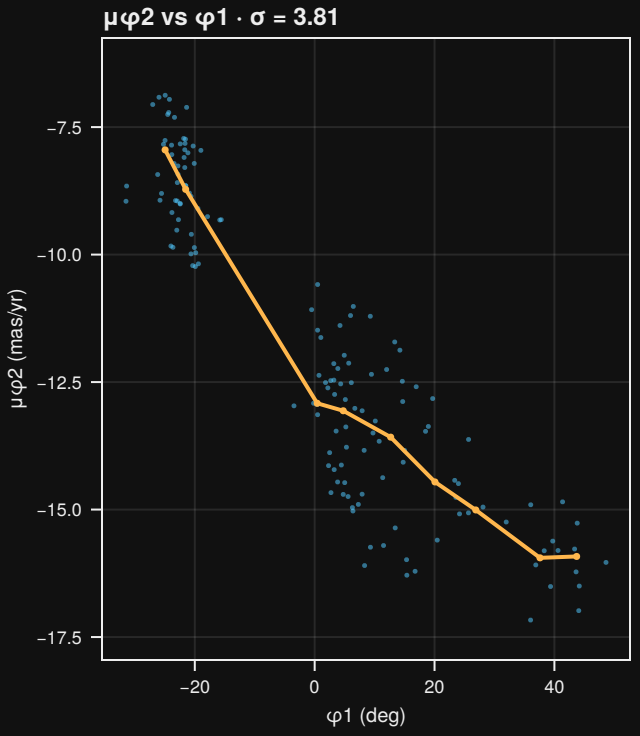


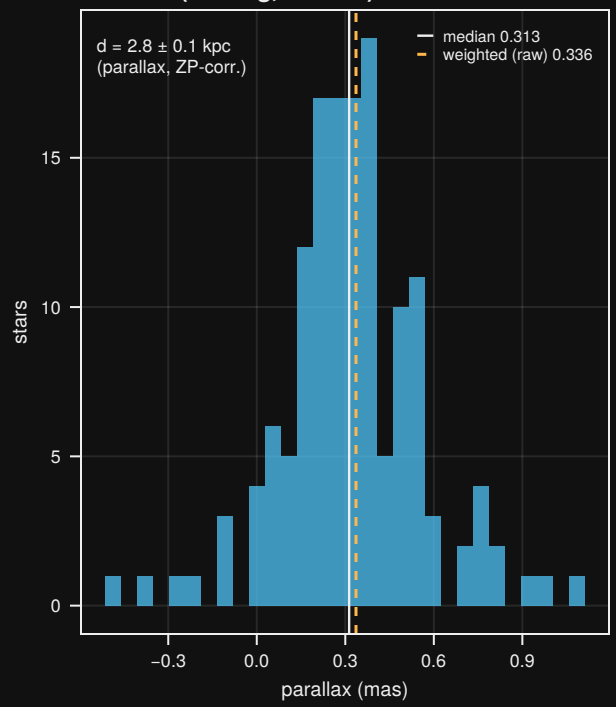
New-6

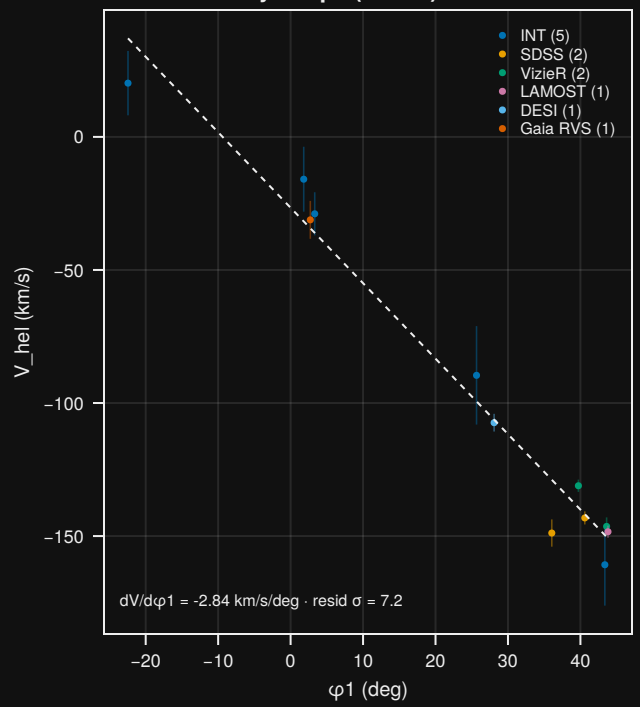
Kinematics · proper motion · parallax · radial velocity

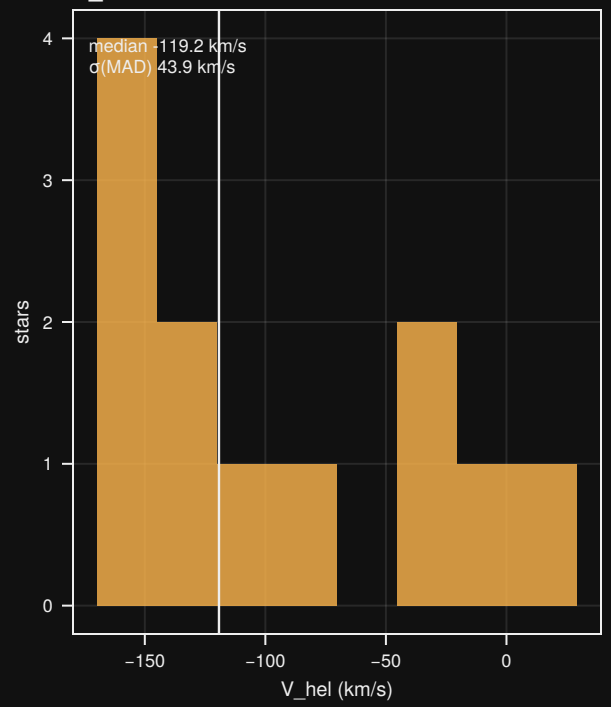






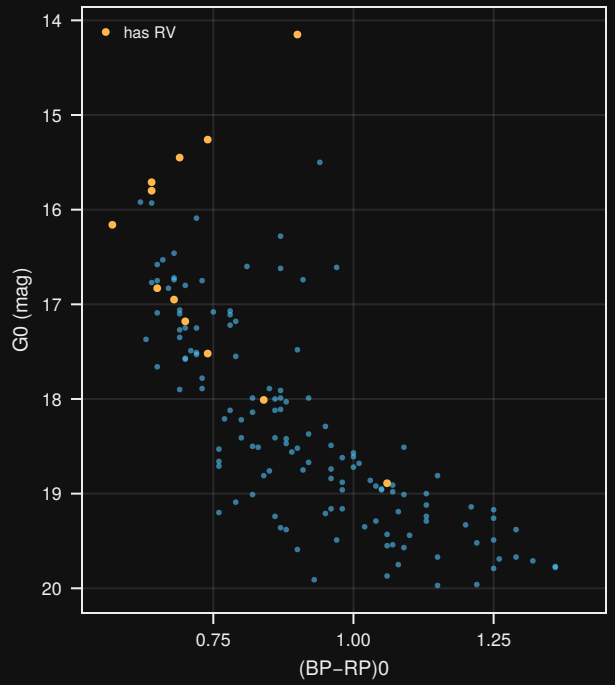




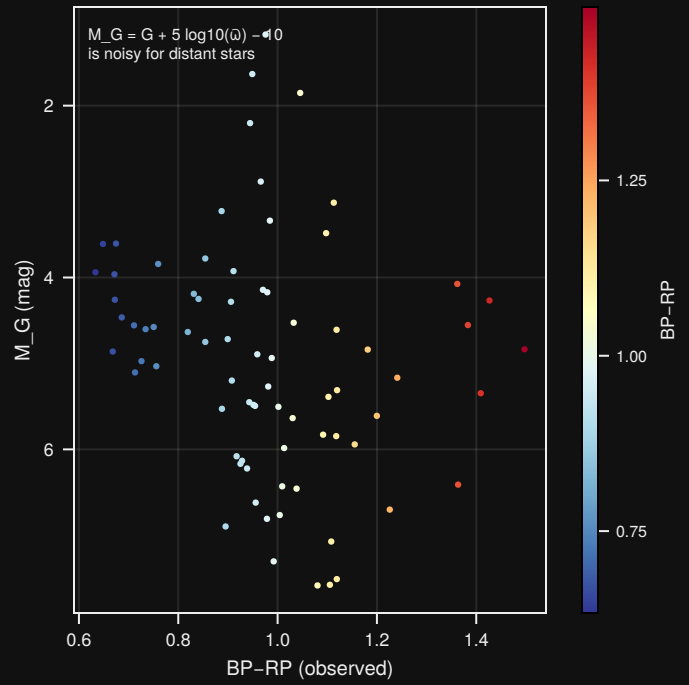


New-6

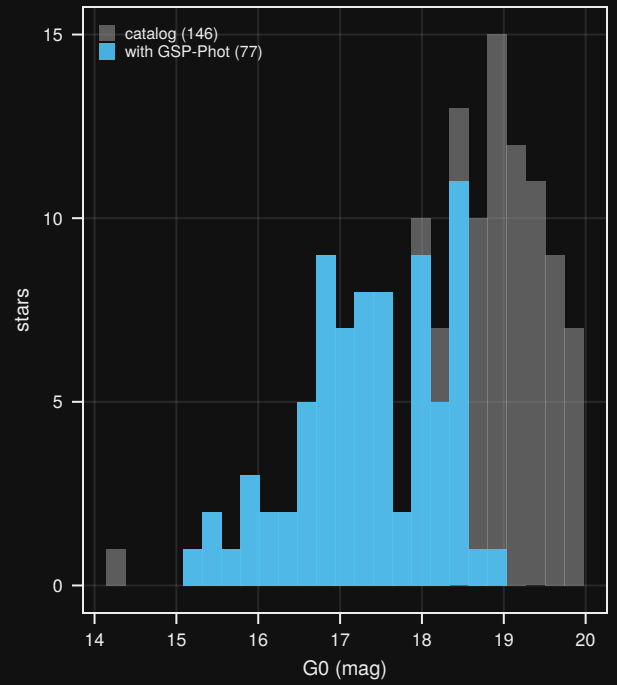
CMD · dereddened (n = 146)



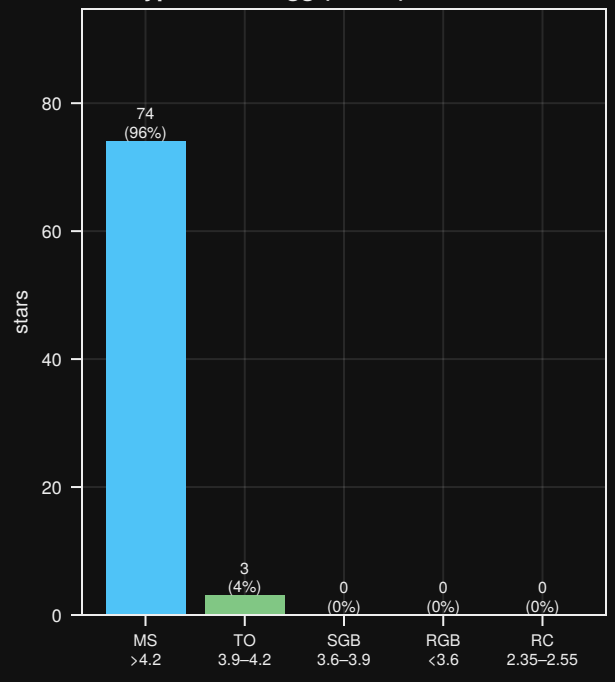
HR diagram · ϖ>0 (n = 76/77)



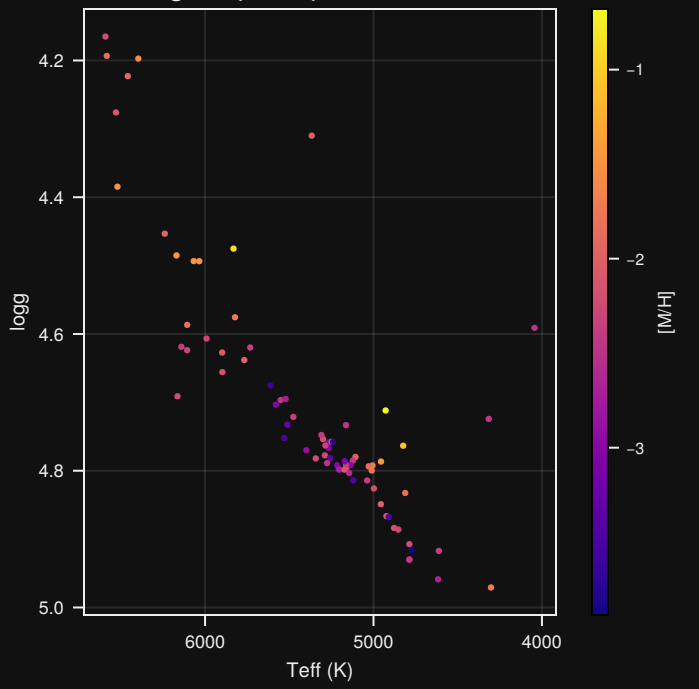
G0 & GSP-Phot coverage



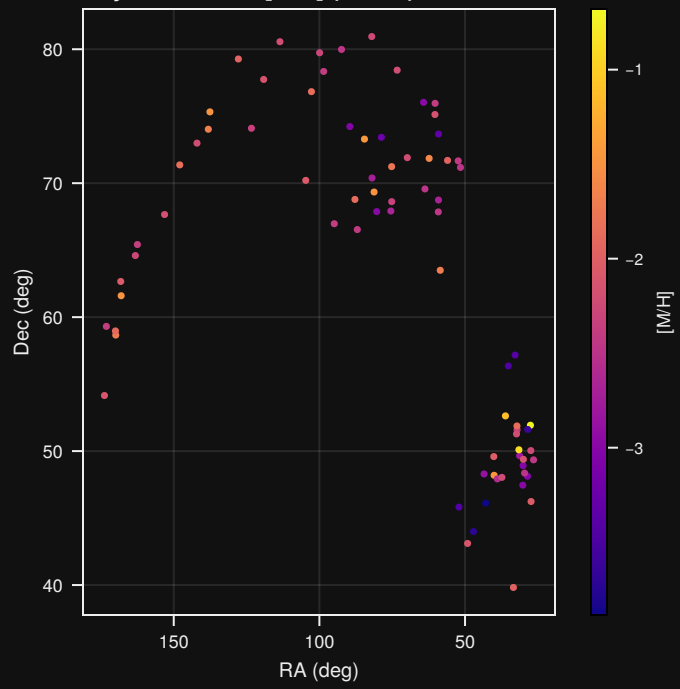
Star types from logg (n = 77)



Kiel diagram (n = 77)



Sky · GSP-Phot [M/H] (n = 77)

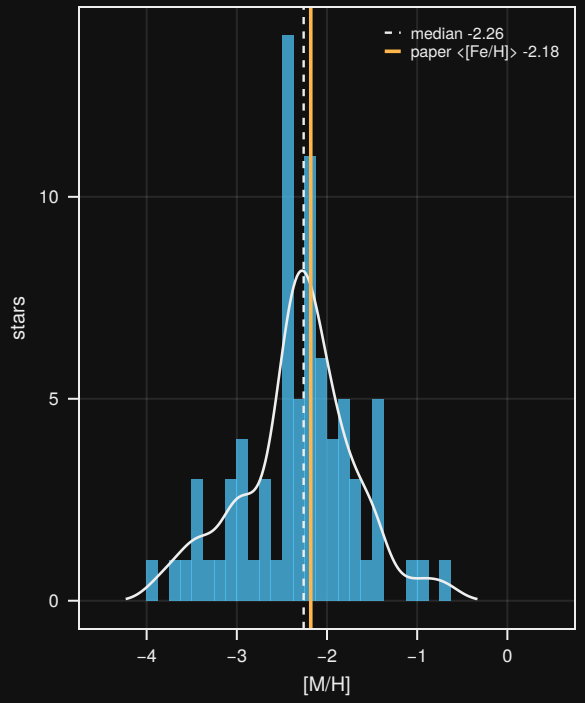


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

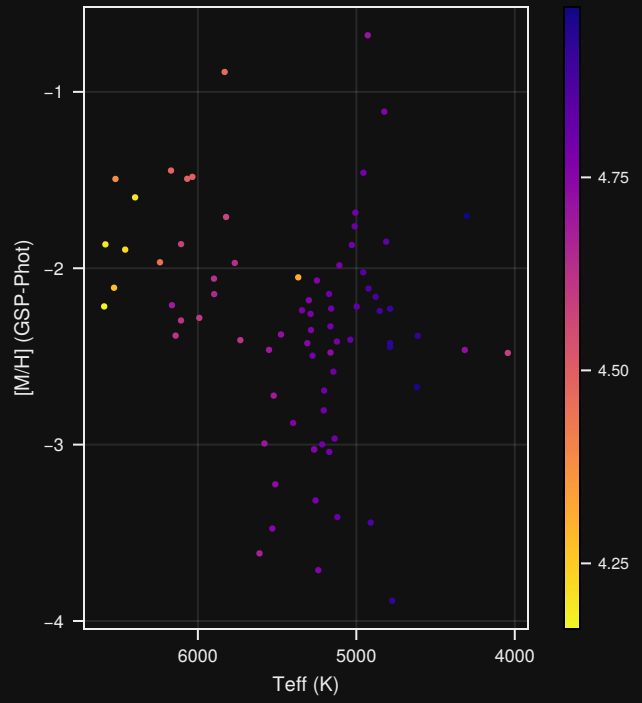
New-6

Metallicity · age · astrometric quality

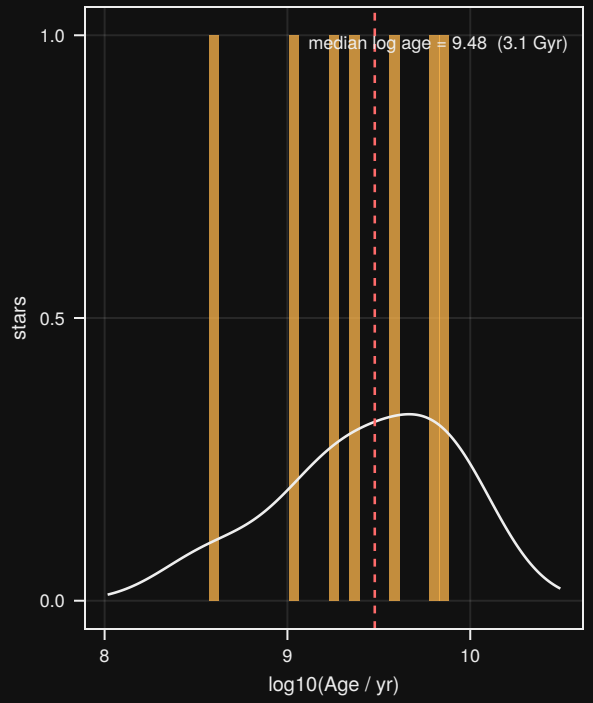
[M/H] GSP-Phot (n = 77)



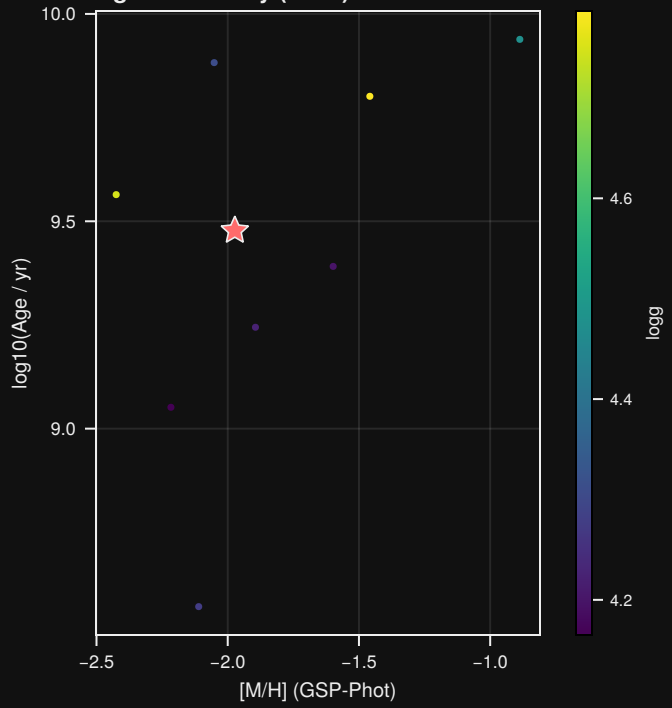
Teff vs [M/H] (n = 77)



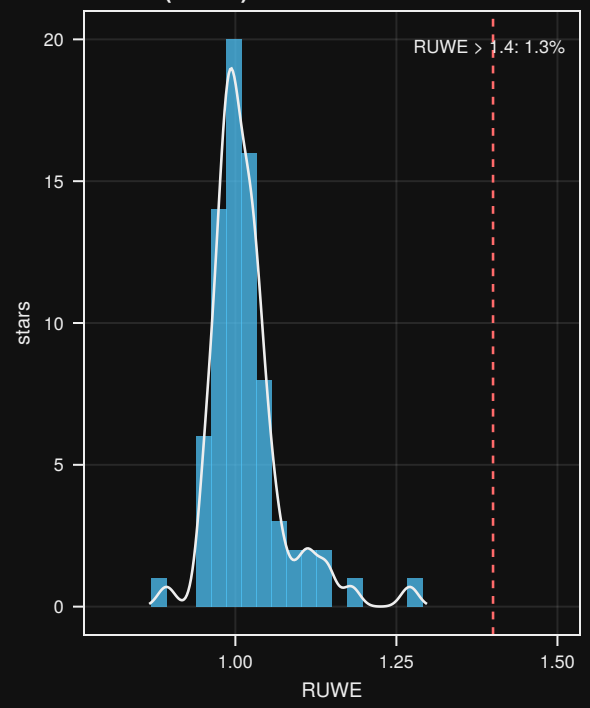
FLAME age (n = 8)



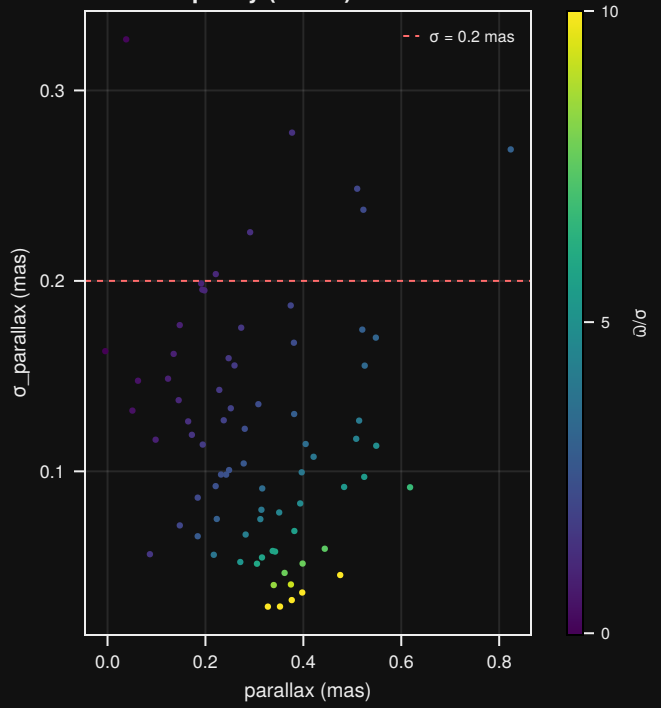
Age–metallicity (n = 8)



RUWE (n = 76)

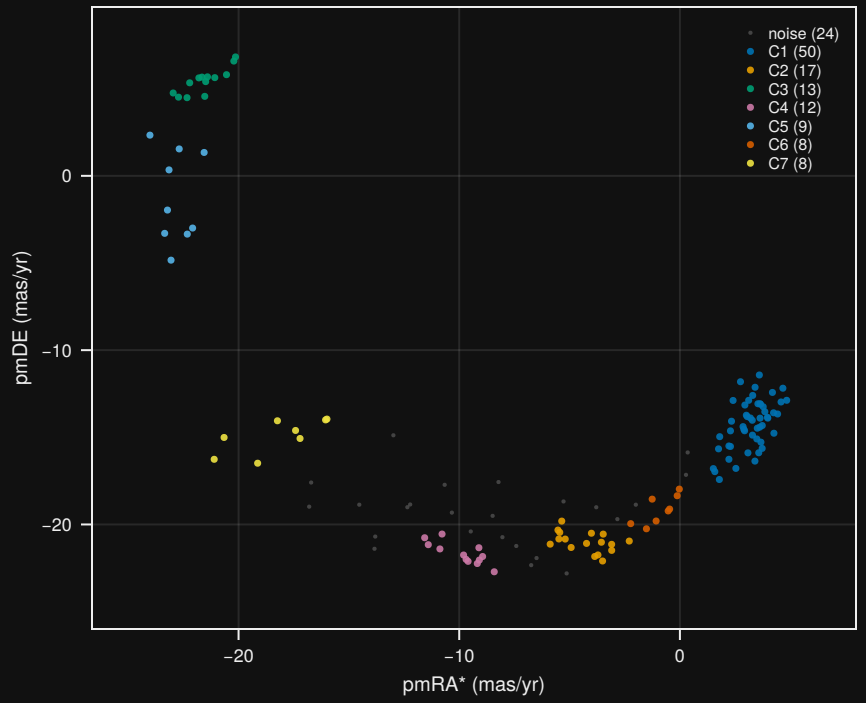


Parallax quality (n = 77)

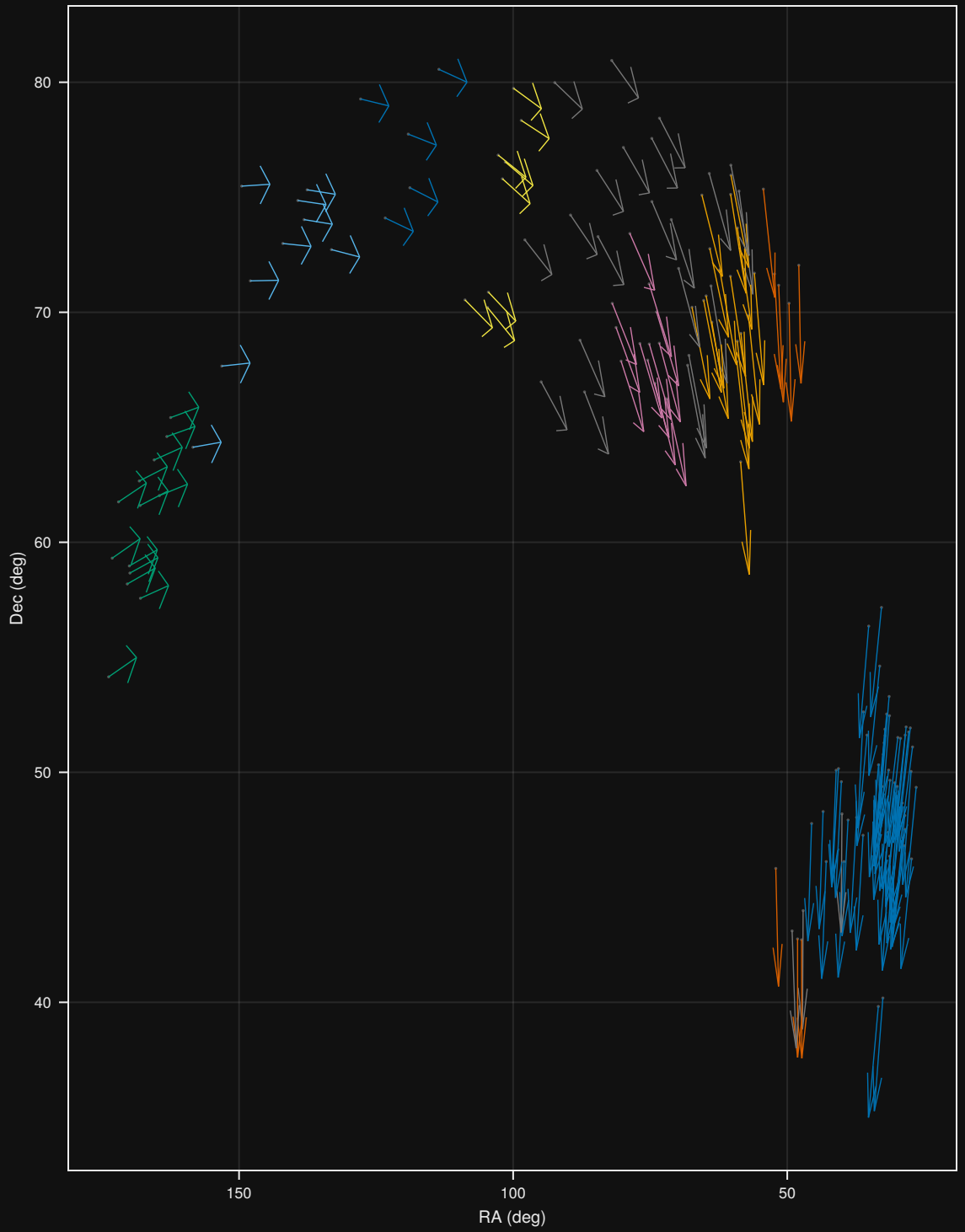


New-6

PM clustering · HDBSCAN · 8 cluster(s) · noise 16%



Proper-motion directions (arrows) coloured by cluster



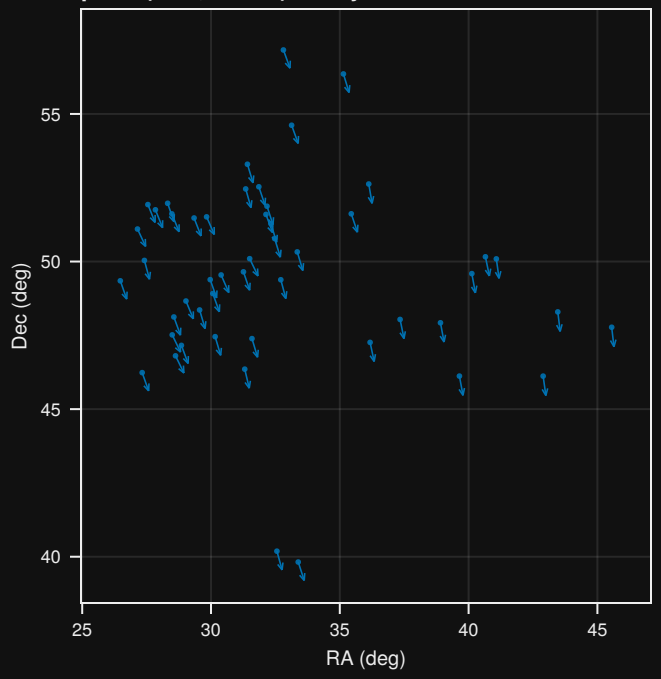
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	24	-8.35	-19.01	5.83	-2.26	0
C1	50	3.35	-14.05	0.65	-2.43	1
C2	17	-4.01	-21.02	1.36	-2.41	2
C3	13	-21.53	5.62	1.01	-2.08	6
C4	12	-9.64	-21.80	0.93	-2.67	1
C5	9	-23.06	-1.96	0.55	-1.85	2
C6	8	-0.80	-19.17	0.84	-2.48	0
C7	8	-17.83	-14.81	2.28	-2.18	0
C8	5	-19.94	-9.24	1.05	-2.18	0

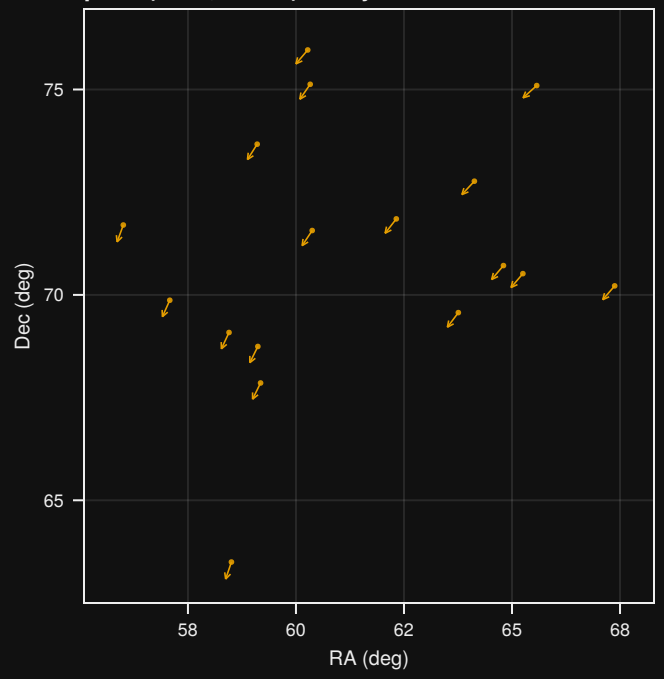
New-6

Per-cluster proper-motion vector maps

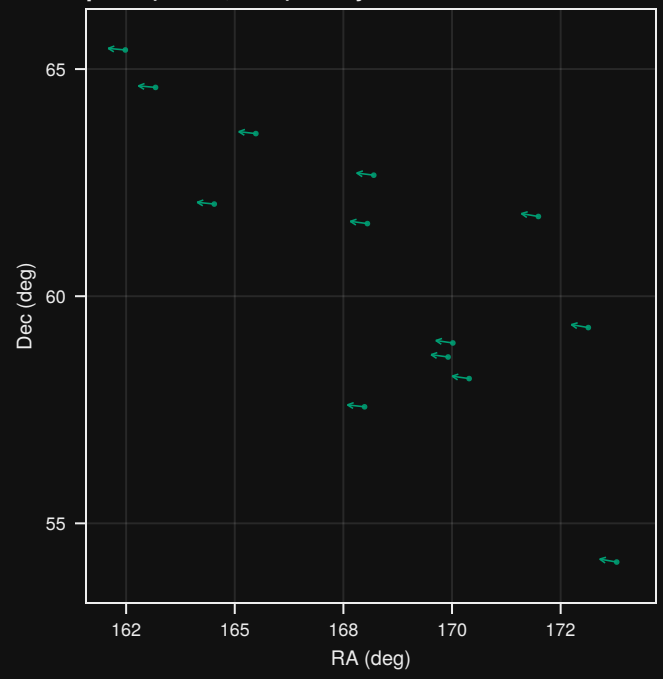
Cluster C1 · n = 50
pm = (3.35, -14.05) mas/yr



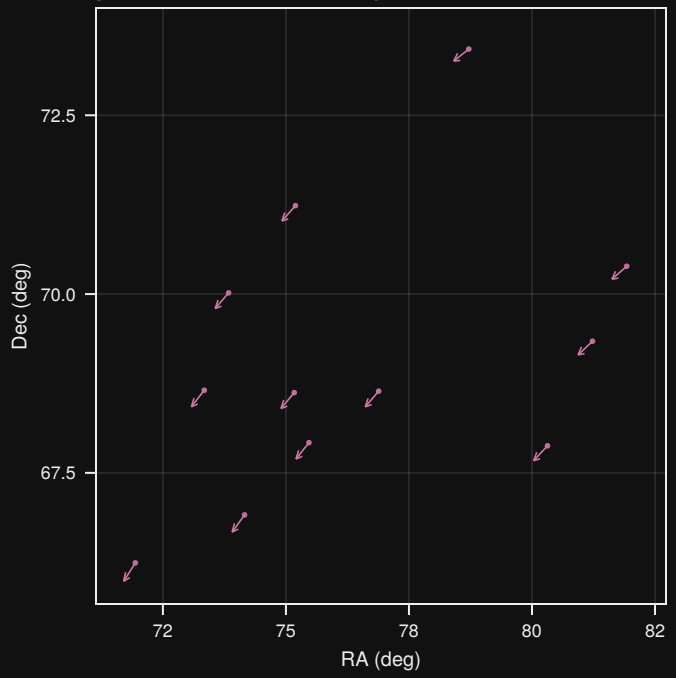
Cluster C2 · n = 17
pm = (-4.01, -21.02) mas/yr



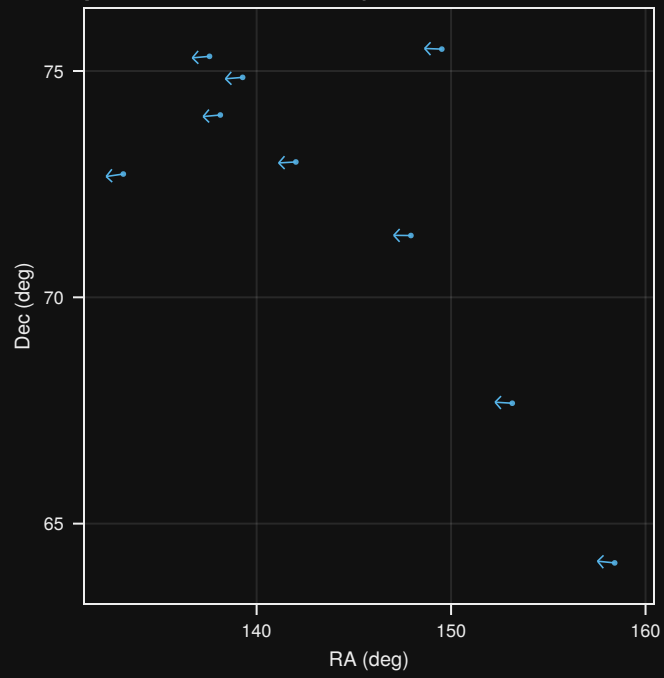
Cluster C3 · n = 13
pm = (-21.53, 5.62) mas/yr



Cluster C4 · n = 12
pm = (-9.64, -21.80) mas/yr



Cluster C5 · n = 9
pm = (-23.06, -1.96) mas/yr



Noise · n = 24

